

# VIVEAK VADIVELKARASAN

## Tech Lead - Distributed Systems & External API Platforms

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### SUMMARY

Engineering leader with 8+ years designing and shipping distributed backend platforms and external-facing APIs that third parties build businesses on. Currently Tech Lead for External APIs at a global travel-experiences company, owning architecture, delivery and engineering standards for its integration platform. Track record of hard migrations done well - serverless-to-Kubernetes re-platforms, organisation-wide containerisation and zero-disruption platform cutovers. Design-first and AI-accelerated: I use AI coding agents and agentic workflows as a force multiplier for architecture, delivery and code review, and coach teams to do the same. Outside work I run an indie product studio, shipping my own products end-to-end.

### SELECTED ACHIEVEMENTS

- **Webhook delivery platform** - Designed and built the event-driven delivery backbone for all external client integrations, processing ~1M events/day: signed payloads, per-subscription circuit breakers, rate limiting, retries and DLQ on AWS EventBridge and BullMQ/Redis.
- **Serverless-to-Kubernetes re-platform** - Re-architected a high-volume data service from AWS Lambdas into a long-running service on EKS - no cold starts or timeout ceilings, fully observable and resumable.
- **AI-accelerated engineering practice** - Led team-wide adoption of AI coding agents and agentic workflows across design, delivery and code review, raising throughput without lowering standards.
- **Self-returns automation** - Shipped an AWS-driven automated returns journey for a global e-commerce brand, cutting customer-service contact volume by ~30%.
- **EC2-to-Kubernetes migration** - Led containerisation for a 300+ engineer organisation - faster deploys, measurable AWS cost reduction.

### SKILLS

**Backend & APIs:** NestJS, Node.js, TypeScript, Java, Python, RESTful APIs, OpenAPI, microservices, event-driven architecture, webhooks, OAuth2/OIDC

**Cloud & Platform:** AWS (EKS, ECS, Lambda, EventBridge, SQS, S3, CloudWatch), Kubernetes, Docker, Terraform, CI/CD, Firebase, GCP, Vercel

**Data & Messaging:** Apache Kafka, BullMQ, Redis, PostgreSQL, Databricks

**Architecture:** Distributed systems, outbox pattern, idempotency, circuit breakers, DLQ, rate limiting, caching semantics, observability (Datadog)

**AI-Accelerated Engineering:** Claude Code, agentic workflows, AI-assisted architecture, development and code review

**Product & Leadership:** Flutter, Stripe, PWAs, Playwright E2E; mentoring, engineering standards, Agile/Scrum, stakeholder management

### PROFESSIONAL EXPERIENCE

#### Collinson

London, UK | Jul 2025 – Present

#### Tech Lead, External APIs

Technical lead for a team of 9 engineers building the external API platform powering lounge reservations, availability and prebooking - the integration surface for 5 major third-party clients, with more being onboarded.

- Designed and built an **event-driven webhook delivery platform** processing ~1M events/day on AWS EventBridge and BullMQ/Redis - HMAC-signed payloads, per-subscription circuit breakers, rate limiting, exponential-backoff retries and a Postgres-backed DLQ.
- **Re-platformed a high-volume data service from AWS Lambdas to EKS**, removing timeout and cold-start constraints and adding observability and back-pressure handling.
- Authored the **push-to-pull integration strategy** (operation-status endpoints with Retry-After / ETag semantics) and its four-phase rollout, cutting client integration cost.
- Own the shared **authentication and authorisation tooling** used across the external APIs estate.
- Set platform standards - outbox publishing, idempotent mutations, correlation-ID tracing, Datadog observability - and lead engineering practice, including team adoption of **AI-assisted development and agentic workflows**.

**Manager, Technology Consulting** (promoted through Engineer / Senior Engineer / Lead)**Technical Lead - LEGO, Customer Data Platform**

Jul 2024 – Jul 2025

- Led integration of Databricks as a Customer Data Platform for LEGO.com, centralising user data to power key metrics, persona modelling and market segmentation.
- Architected the AWS pipeline consolidating source streams into Salesforce CRM; recovered a delayed initiative and brought delivery back on schedule.

**Technical Lead Engineer - UK Government**

Feb 2023 – Jul 2024

- Turned around an underperforming delivery team - reset cadence, closed deadline gaps and raised throughput while directing MVP scope and roadmap.
- Introduced the client's first asynchronous service journey and provided architecture guidance that strengthened the relationship and secured follow-on engagement scope.

**Lead Engineer - LEGO, Large-Scale SFMC Migration**

Jun 2021 – Dec 2022

- Led four engineers migrating LEGO's communication channels from Oracle Crowdtwist to Salesforce Marketing Cloud with zero customer-visible disruption.
- Built supporting AWS integrations for scale and reliability of the new marketing pipeline; ran technical stakeholder sessions through cutover.

**Full-Stack Developer - LEGO**

Jun 2020 – Jun 2021

- Designed and shipped the AWS-driven automated self-returns system on LEGO.com, cutting customer-service contact volume by ~30%.

**Lead Engineer - Dyson**

Nov 2019 – May 2020

- Led migration from EC2 to Kubernetes on AWS for a 300+ engineer organisation - auto-scaling, faster deploys, measurable cost reduction; built automated UAT and performance environments.

**Frontend Engineer - Dyson**

Jun 2018 – Nov 2019

- Built reusable AEM components rolled out across Dyson's European e-commerce estate.

**Gentrack - Software Engineer** (Mar 2018 – Jun 2018) - Backend invoicing solutions in Java.

**Accenture - Application Developer** (Jul 2017 – Mar 2018) - Drove HMRC's automated test coverage from ~25% to 75%+, converting a graduate offer into a permanent role.

**INDEPENDENT PRODUCTS - AADHIPIXELS (FOUNDER)**

Indie product studio where I design, build, ship and operate products end-to-end - design-first and AI-accelerated.

- **TheCocoCard** - Digital loyalty platform for local businesses: business back-office, consumer PWA, Flutter app, Firebase backends and cross-app Playwright E2E, deployed on Vercel and Firebase.
- **Fixabee** - Local-services marketplace in Flutter and Firebase: bookings, provider onboarding with Stripe payouts, postcode-based discovery, referrals and full dev/test/prod release environments.
- **Rapid prototypes** - DoDay, ClimaGo, ShouldIgo, GrowMeAForest, Buy My Postcode - a habit of taking ideas from concept to working product fast.

**EARLY CAREER**

- **Tullett Prebon**, Software Engineer Intern (2016) - Built a market-data trading platform used by traders during the UK EU referendum.
- **Accenture**, Industrial Placement (2015 – 2016) - Overhauled automated test strategy, lifting coverage from <25% to 75%+.

**EDUCATION & CERTIFICATIONS**

- **BSc (Hons) Computer Science** - City, University of London, 2017
- **AWS Certified Solutions Architect**